Juan David Muñoz-Bolaños

Optical System Engineer

Rechenauerstraße 42, Innsbruck, Österreich (Tirol)

+43 676 9115145
juan.munoz@i-med.ac.at | LinkedIn Profile

Summary

Physics Engineer with deep expertise in **Optics, Laser Systems, and Machine Vision**. Experienced in delivering industrial Halcon pipelines for series production and building complex adaptive optics metrology systems. Currently finishing a PhD focused on high-precision hardware-software integration.

Professional Experience

Mar 2018 to Mar 2019 **INTECOL S.A.S.** Medellín, Colombia
Junior Machine Vision System Engineer

Machine Vision Pipelines: Developed and commissioned Halcon-based inspection systems for high-speed glass lines (labels, sealing, liquid level). Handled custom illumination design and hardware integration.

Robotic Vision: Implemented Cognex VisionPro for automated part recognition, enabling robotic flame treatment of automotive components (Renault).

Jan 2022 to Present **Medical University of Innsbruck** Innsbruck, Austria
PhD Researcher, Adaptive Optics & Data Processing

Optics & Metrology: Built a fully automated wavefront sensor stack using C++/Python/LabVIEW; integrated adaptive optics for diffraction-limited imaging (ACS Photonics '25).

System Integration: Full-stack development from hardware interfacing to real-time wave-field control and precision data processing in Python.

Jun 2020 to Dec 2021 **Leibniz IPHT** Jena, Germany
M.Sc. Researcher | Hyperspectral AI & Photonics

AI & Spectroscopy: Developed the SpectraMap Python package for hyperspectral Raman image analysis. Leveraged clustering and ML algorithms for automated cancer detection in tissue samples (Applied Sciences '24).

Spectrometer Prototyping: Led the optical design and build of a 1064 nm Raman spectrometer.

Technical Competencies

Image Processing & Computer Vision: Halcon (MVTec), Cognex VisionPro, OpenCV; object detection, segmentation, defect detection; illumination design (bright-field, dark-field, structured light).

AI & Machine Learning: Supervised ML on image and spectral data (Python, NumPy, scikit-learn); clustering and classification models for automated defect detection and object recognition.

Software Development: Python (advanced) · Halcon (intermediate) · C++ (intermediate / OpenCV) · LabVIEW/FPGA · Git · MATLAB; knowledge of C#/.NET concepts.

3D Metrology & Simulation: Phase retrieval, Zernike decomposition, wave-field modeling in Python; ZEMAX OpticStudio tolerance analysis.

Industrial Integration: Hardware-software integration of camera systems, laser modules, and positioning stages; high-precision alignment of optical measurement systems; commissioning on production lines.

Laser Systems Alignment: Expertise in integration and alignment of high-power laser systems (fs, CW, 1064 nm); maintenance of precision opto-mechatronic setups and EH&S compliance.

Education

Jan 2022 to Present	**Medical University of Innsbruck** *PhD in Adaptive Optics & Microscopy*	Austria
2019 to 2021	**Friedrich Schiller University Jena** *M.Sc. in Photonics*	Germany
2012 to 2018	**University of Cauca** *B.Sc. in Physics Engineering*	Colombia

Publications & Awards

Awards: ASML & ZEISS Summer Schools (Selected); COLFUTURO Grant 2026 Awardee.

Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).

Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).

Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

Languages

English (Professional) · German (B2 Certificate | High Intermediate) · Spanish (Native)

Soft Skills & Hobbies

Problem Solving: Resolved complex, cross-disciplinary challenges resulting in 3 peer-reviewed publications.

Adaptability: Built a global career path across Colombia, Germany, and Austria with strong networking skills.

Hobbies: Mountain biking, Bachata dancing, harmonica, learning German.

References

Prof. Monika Ritsch-Marte — Medical University of Innsbruck
Head of the Institute of Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft — Leibniz-IPHT Jena
Spectroscopy Group Leader
christoph.kraft@leibniz-ipht.de